

CS-AML

Software Requirements Specification (SRS)

Version 0.1.4

Approved Internal Specification Baseline

2026-10-10 · tag v0.1.4-spec

Civil Society Anti-Money Laundering & Financial Intelligence Framework

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Version: 0.1.4 — Approved Internal Specification Baseline (2026-10-10, tag v0.1.4-spec). Supersedes v0.1.3 (2026-10-09, tag v0.1.3-spec). [v0.1.4]

Supersedes: CS-AML Software Requirements Specification (SRS) v0.1. The DOCX/PDF files in this repository are the unchanged v0.1 baseline (legacy); this Markdown file is the canonical source.

Validation: approved by the product owner as the internal specification baseline on 2026-10-08 (v0.1.1) and 2026-10-09 (v0.1.2 and v0.1.3) and 2026-10-10 (v0.1.4; decision register and release gates in CHANGELOG.md). The v0.1.2, v0.1.3 and v0.1.4 changes come from change requests raised while implementing increments I1-I4, I5-I7 and the post-MVP follow-ups; this is not an independent audit. Acceptance criteria in this document are targets, not evidence that tests have passed.

CS-AML is not an external standard or certification. References to FATF, Wolfsberg, PPATK, UNODC or other bodies do not imply their endorsement.

Changes in 0.1.1: see CHANGELOG.md at the repository root (audit findings A01-A16).

Changes in 0.1.2: change requests CR-I1-01...CR-I4-14 approved by the product owner on 2026-10-09 (CHANGELOG.md, section v0.1.2). Each change is tagged *[v0.1.2 · CR-xx-yy]*.

Changes in 0.1.3: change requests CR-I5-01...CR-I7-08 approved by the product owner on 2026-10-09 (CHANGELOG.md, section v0.1.3). Each change is tagged *[v0.1.3 · CR-xx-yy]*.

Changes in 0.1.4: change requests CR-N-01...CR-N-14 approved by the product owner on 2026-10-10 (CHANGELOG.md, section v0.1.4). Each change is tagged *[v0.1.4 · CR-N-xx]*.

Document status

Proposed normative software baseline for MVP 0.1 implementation (draft for review). This SRS translates the PRD (draft for review), Product & Feature Specification, Data Model Specification, Control Implementation Guide, Investigation Methodology, and Technology Architecture into testable software requirements. [v0.1.1 · A01]

Civil Society Financial Intelligence / AML Investigation Platform

Document Control

Document	CS-AML Software Requirements Specification
Version	0.1.4 [v0.1.4]
Status	Approved Internal Specification Baseline (2026-10-10, tag v0.1.4-spec) [v0.1.1 · A01]
Primary product scope	MVP 0.1
Audience	Product, engineering, QA, security, data, reviewers, governance
Upstream documents	CS-AML PRD v0.1.1; Product & Feature Specification v0.1.1; Data Model Specification v0.1.4; Technology Architecture v0.1.1; Control Implementation Guide v0.1.4; Investigation Methodology v0.1.4 (Markdown, Documents/*_v0.1.1.md)
Normative terms	SHALL / MUST = mandatory; SHOULD = recommended; MAY = optional

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1. Purpose and Scope

This SRS specifies the software behaviour, data obligations, interfaces, controls, quality attributes, and verification conditions required to implement CS-AML MVP 0.1 as a secure system of record for civil-society financial-intelligence investigations.

Primary acceptance objective

A small two-analyst team SHALL be able to open a sensitive case, ingest and preserve evidence, resolve entities, map relationships/assets/events/value flows, test competing hypotheses, assess typology consistency and confidence, complete independent review, and export an approved intelligence product while retaining provenance and audit history.

1.1 In scope

- Case and workflow management
- Source/evidence/provenance management
- Document ingestion
- Entity registry and reversible entity resolution
- Relationships, ownership, control, and assets

- Events and timeline
- Value-flow modelling
- Typology and indicator analysis
- Hypothesis, gap, confidence, and assessment management
- Search and graph exploration
- Intelligence products, review, dissemination and sharing logs
- IAM, protected-source compartment, audit, retention, backup and restore

1.2 Out of scope for MVP 0.1

- Automated AML transaction monitoring against banking core systems
- Autonomous guilt determination or law-enforcement decisioning
- Continuous web-scale scraping
- Cross-organisation federation
- Advanced blockchain tracing
- Automated sanctions/PEP screening as a mandatory workflow
- Advanced machine-learning risk scoring
- Multi-tenant SaaS operation

2. Product and System Context

CS-AML is positioned as a civil-society financial-intelligence and investigation platform, not a bank compliance engine. The software converts structured investigative work into reproducible intelligence while preserving uncertainty and provenance.

2.1 Product axiom

Axiom

Technology SHALL preserve analytical uncertainty rather than erase it. Source, evidence, claim, fact, indicator, hypothesis, assessment, and intelligence product remain distinguishable in storage, APIs, UI, graph views, automation, and exports.

2.2 Upstream analytical chain

| SOURCE → EVIDENCE → CLAIM/FACT → INDICATOR → HYPOTHESIS → ASSESSMENT → INTELLIGENCE PRODUCT

2.3 Mandatory value-flow classes

| DIRECT | DOCUMENTED | RECONSTRUCTED | HYPOTHETICAL

These are the `flow_class` wire values. All controlled enumerations on the wire (database values, API payloads, exports) use UPPER_SNAKE_CASE and derive from the Data Model Specification v0.1.4 Annex A registry; display labels are separate and translatable. [v0.1.1 · A09]

3. Definitions and Conventions

Term	Meaning	Software implication
Canonical object	Authoritative record of investigative state	Must be durable, versioned where material, access-controlled, and auditable

Term	Meaning	Software implication
Derived projection	Rebuildable search/graph/cache/ AI representation	Must never become sole source of truth
Material action	Action capable of changing analytical meaning, access, review status, or dissemination	Must be auditable
Protected source	Human/source identity requiring compartmentalisation	Identity access separate from routine evidence access
High-impact product	Product containing potentially harmful adverse findings or public attribution	Requires independent review and dissemination approval
MVP	Minimum release satisfying end-to-end case completion	All P0 requirements required unless explicitly waived; a defect against a non-waivable invariant (§21) can never be waived [v0.1.1 · A16]
Information classification	Five ordered levels (least → most restrictive), wire values PUBLIC, INTERNAL, SENSITIVE, RESTRICTED, SOURCE_PROTECTED (display: Public, Internal, Sensitive, Restricted, Source-protected), authoritative per Data Model Specification v0.1.4 §16	Access labels (purpose, jurisdiction, embargo, legal-review, compartment, etc.) are additive; the most restrictive applicable level plus all labels apply; derived objects/exports inherit the highest input classification unless a recorded reviewer downgrade decision exists; unknown or missing classification fails closed (deny and flag for classification). Framework v0.1 "Highly Restricted" is never auto-mapped to SOURCE_PROTECTED [v0.1.1 · A08]. A declared classification below the inputs' maximum (or labels missing an input label) is rejected (422), never silently raised; when an input is later upgraded, derived objects are flagged for re-review and their classification is not raised automatically [v0.1.2 · CR-I2-05]

4. System Boundary and Actors

4.1 Primary actors

Actor	Responsibility
Investigator	Creates and develops case material
Case Owner	Accountable for scope, gates, and dissemination decisions
Analyst	Performs entity/value-flow/typology/hypothesis analysis
Reviewer	Independent quality review
Evidence Custodian	Preserves originals, integrity, and evidence lineage
Data Steward	Approves significant entity merge/unmerge and data-quality changes
Source Handler	Controls protected-source identity
Platform/Security Admin	Operates IAM, configuration, backup, monitoring
Auditor	Reviews controls and audit history

4.2 Trust boundaries

- User browser/client
- Application/API policy layer
- Canonical database
- Evidence object store
- Protected-source compartment
- Search/graph projections
- External integrations
- Backup/restore environment
- Administrative control plane

5. Architectural Constraints

- Canonical data and evidence remain independent from rebuildable search/graph projections.
- Original evidence is immutable to routine analyst workflows; transformations create derivatives.
- Entity merge decisions are reversible and must preserve history.
- All material adverse analytical claims remain traceable to evidence.
- Authorization is enforced at API/service layer, not only UI.
- High-impact dissemination requires accountable human authorization.
- AI and automation may assist but cannot silently turn generated output into material facts. *[v0.1.1 · A10]*

6. Functional Requirements

6.1 Case and Workflow

SRS-FR-CASE-001 — Case creation

Requirement	The system SHALL create a stable Case ID and require title, accountable owner, purpose, investigation question, sensitivity/classification, and status before a case becomes Active. At least one investigation question is a precondition of case activation (gate), not of case creation: a DRAFT case MAY have none, and the Case representation returns an empty list until the charter defines one. <i>[v0.1.2 · CR-I1-01]</i> <code>risk_rating</code> uses the registry enum <code>risk_rating</code> (LOW, MEDIUM, HIGH, CRITICAL) and <code>closure_reason</code> the registry enum <code>closure_reason</code> . <i>[v0.1.2 · CR-I1-08]</i> Only the case LEAD MAY change a case's classification or access labels, and only upwards (higher level, added labels); downgrades and label removal require a recorded reviewer decision (workflow delivered with review, I6) and SHALL be rejected (403) until then. <i>[v0.1.2 · CR-I1-09]</i>
Rationale	Prevents orphaned or undefined investigations.
Verification	Create incomplete case and verify activation is blocked; complete required fields and verify activation succeeds. Create a case without questions and verify it is accepted as DRAFT with <code>investigation_questions: []</code> ; attempt a classification downgrade as LEAD and an upgrade as non-LEAD and verify 403. <i>[v0.1.2 · CR-I1-01, CR-I1-09]</i>
Priority	MVP / P0
Traceability	F-CASE-001; GOV-01; CAS-01

SRS-FR-CASE-002 — Investigation Charter

Requirement	The system SHALL version the Investigation Charter, including question, scope, exclusions, jurisdictions, period, authorised/prohibited collection methods, risks, and intended outputs.
Rationale	Controls scope and mission creep.
Verification	Edit charter twice and verify prior version remains retrievable with actor/time. Verify a second version without <code>change_rationale</code> is refused (422) and <code>GET /cases/{caseId}/charter/versions/{version}</code> returns each version. <i>[v0.1.4 · CR-N-01]</i>
Priority	MVP / P0
Traceability	F-CASE-002; CAS-01; PRI-01

SRS-FR-CASE-003 — Lifecycle gates

Requirement	The system SHALL support configurable G0-G6 lifecycle gates and SHALL prevent controlled actions when required approval is absent. In v0.1.4 the controlled actions are the case-lifecycle transitions: the case status changes only through an approved gate (G0 DRAFT → AUTHORIZED, G1 AUTHORIZED → ACTIVE, G4 ACTIVE → REVIEW, G6 ACTIVE / REVIEW → CLOSED); collection, assessment finalisation and dissemination keep their own controls. The decider SHALL NOT be the submitter; for high-impact cases (risk HIGH / CRITICAL or classification RESTRICTED / SOURCE_PROTECTED) the G0 / G1 decider SHALL NOT have authored the charter version under approval. Readiness criteria are checked where they are data (API §12) and otherwise left to reviewer judgement. Gate configuration per organisation or risk tier, MONITORING and reopening are deferred to v0.2. <i>[v0.1.4 · CR-N-03, CR-N-05, CR-N-06, CR-N-07]</i>
Rationale	Enforces stage-gated investigation methodology.
Verification	Attempt high-impact transition without approval and verify denial; approve independently and retry.
Priority	MVP / P0
Traceability	F-CASE-003; CAS-02; QUA-01

SRS-FR-CASE-004 — Tasks and assignments

Requirement	The system SHALL support case-scoped tasks with owner, due date, type, status, dependencies, and immutable completion history.
Rationale	Coordinates investigative work.
Verification	Create, reassign, complete task and verify audit trail. Verify DONE is refused while a dependency is open (409), a closed task cannot change (409) and a non-member assignee is refused (422). <i>[v0.1.4 · CR-N-04]</i>
Priority	MVP / P0
Traceability	F-CASE-004

SRS-FR-CASE-005 — Case activity timeline

Requirement	The system SHALL display material case actions in chronological order with actor, timestamp, object, and action type. Only events the viewer may see are shown (no authorization / authentication / security events). Filtering by actor, action and object is client-side in v0.1.4; server-side filter parameters and an auditor view are deferred to v0.2. <i>[v0.1.4 · CR-N-08]</i>
Rationale	Supports reconstructability.
Verification	Perform material actions and verify ordered visibility.
Priority	MVP / P0
Traceability	F-CASE-005; AUD-01

6.2 Source, Evidence and Documents

SRS-FR-EVD-001 — Source register

Requirement	The system SHALL register material sources with origin, type, publisher/owner where known, access method, access timestamp, URL/location, reliability, legal/access note, and archival reference where available.
Rationale	Maintains provenance.
Verification	Create source and verify mandatory provenance validation.
Priority	MVP / P0
Traceability	F-EVD-001; SRC-01

SRS-FR-EVD-002 — Original preservation

Requirement	The system SHALL preserve original evidence separately from working/derived artefacts and SHALL prevent routine analysts from modifying the original object.
Rationale	Protects evidence integrity.
Verification	Upload file, attempt overwrite as analyst, verify denial and new-version workflow.
Priority	MVP / P0
Traceability	F-EVD-002; EVD-01

SRS-FR-EVD-003 — Cryptographic integrity

Requirement	The system SHALL compute SHA-256 or stronger approved hash for critical uploaded evidence and SHALL support recomputation/verification.
Rationale	Detects silent change.
Verification	Recompute hash and compare; alter test copy and verify mismatch.
Priority	MVP / P0
Traceability	F-EVD-003; EVD-01

SRS-FR-EVD-004 — Evidence extracts

Requirement	The system SHALL allow page/paragraph/region extracts that remain linked to parent evidence and a precise source location.
Rationale	Enables citations and review.
Verification	Create extract and navigate back to original location.
Priority	MVP / P0
Traceability	F-EVD-004; EVD-02

SRS-FR-EVD-005 — Derivative lineage

Requirement	The system SHALL record parent evidence, transformation type, tool/version where material, creator, and timestamp for OCR, translation, crop, parse, or derived dataset. OCR, text extraction and machine translation produce a DerivedText (Data Model §7.7) plus a derivative EvidenceItem with derivation_type OCR_TEXT, TEXT_EXTRACTION or MACHINE_TRANSLATION. [v0.1.4 · CR-N-09, CR-N-10]
Rationale	Preserves transformation lineage.
Verification	Create derivative and verify reverse trace to original.
Priority	MVP / P0
Traceability	F-EVD-005; EVD-02

SRS-FR-EVD-006 — Reliability and credibility

Requirement	The system SHALL store source reliability independently from information credibility and SHALL not reuse one value for both dimensions. Information credibility uses the registry enum credibility_grade (named values with digit wire codes "1"–"6"); source reliability uses grades A–F. [v0.1.2 · CR-I2-08]
Rationale	Prevents false certainty.
Verification	Save different ratings and verify independent display/storage.
Priority	MVP / P0
Traceability	F-EVD-006; SRC-02; ASM-01

6.2a Claim and Fact Lifecycle [v0.1.1 · A10]

Status of this family

SRS-FR-CLM-001...004 were added in v0.1.1 to close audit finding A10. Approved by product owner, 2026-10-08. A Claim is a permanent record of what a source asserts and is never converted into a Fact; a Fact is a separate object supported by evidence (mandatory), optionally by claims, and by VerificationDecisions. [v0.1.1 · C02] Object definitions follow Data Model Specification v0.1.4 §7.4 Claim and §7.5 Fact.

SRS-FR-CLM-001 — Claim record and attribution

Requirement	The system SHALL record each Claim as a source-attributed assertion linked to its asserting source/person and supporting evidence/extract references, with <code>claim_status</code> ∈ RECORDED, UNDER_REVIEW, CORROBORATED, CONTRADICTED, UNRESOLVED. Analyst conclusions SHALL NOT overwrite the claim's asserted content or attribution. The disputed boolean is retained for compatibility and SHALL be derived (true when status is CONTRADICTED or an open dispute exists).
Rationale	Keeps what a source asserted separate from what analysts concluded.
Verification	Create a claim from an extract; record an analyst conclusion; verify the claim content and attribution are unchanged and disputed is derived from status.
Priority	MVP / P0
Traceability	F-EVD-008; SRC-01; EVD-02

SRS-FR-CLM-002 — Verification decision

Requirement	Every verification outcome on a claim or fact SHALL be stored as a separate, append-only VerificationDecision record (claim decisions through POST <code>/claims/{claimId}/verification-decisions</code> ; fact decisions only atomically with fact creation and the fact commands [v0.1.1 · C01]) containing <code>target_ref</code> (claim or fact), <code>decision</code> (claims: UNDER_REVIEW, CORROBORATED, CONTRADICTED, UNRESOLVED; facts: CREATE, ESTABLISH, DISPUTE, SUPERSEDE) [v0.1.1 · A10], <code>rationale</code> , <code>evidence_refs</code> , <code>decided_by</code> , <code>decided_at</code> , and optional <code>review_ref</code> . Existing decisions SHALL NOT be edited or deleted. Claim decisions MAY be recorded by members with the LEAD, ANALYST or REVIEWER role; claim transitions are RECORDED → UNDER_REVIEW → CORROBORATED / CONTRADICTED / UNRESOLVED, and a decided claim MAY return to UNDER_REVIEW (re-review; history kept); any other transition SHALL be rejected (409). [v0.1.2 · CR-I2-11]
Rationale	Makes every verification judgement reviewable and attributable.
Verification	Record two decisions on one claim; attempt to modify the first and verify rejection; verify both remain retrievable in order with attribution.
Priority	MVP / P0
Traceability	F-EVD-008; AUD-01; QUA-01

SRS-FR-CLM-003 — Fact creation from supporting claims and evidence [v0.1.1 · A10]

Requirement	The system SHALL create a Fact only when it references supporting evidence (supporting_evidence, 1..n, mandatory) and carries a decision rationale; supporting claims (supporting_claim_refs, 0..n) are optional. The system SHALL create the Fact and its CREATE VerificationDecision atomically in one transaction (API Specification §16A); the fact commands establish/dispute/supersede SHALL likewise create their decision atomically. [v0.1.1 · C01] [v0.1.1 · C02] A new Fact SHALL start as PROVISIONAL. Creating a Fact SHALL NOT modify or convert the supporting claims. fact_status ∈ PROVISIONAL, ESTABLISHED, DISPUTED, SUPERSEDED. Investigators/Analysts MAY record claims and propose PROVISIONAL facts; moving a fact to ESTABLISHED SHALL require a Reviewer who is not the proposer, i.e. a principal holding the REVIEWER or LEAD membership role on the fact's case who did not propose the fact (403 otherwise). [v0.1.2 · CR-12-11] Evidence extracts and AI/automation output SHALL NOT become facts except through this path.
Rationale	Prevents a claim or extract from being treated as fact without a reviewable decision.
Verification	Attempt to create a fact without evidence refs or decision rationale and verify rejection (422); create a fact and verify its CREATE decision exists in the same transaction [v0.1.1 · C01]; proposer attempts to establish own fact and is denied; an independent reviewer establishes it and the decision is recorded.
Priority	MVP / P0
Traceability	F-EVD-008; EVD-02; QUA-01

SRS-FR-CLM-004 — Fact revision and dependent flagging

Requirement	Any authorized case member SHALL be able to move a fact to DISPUTED with supporting evidence; SUPERSEDED SHALL require a replacement fact reference (superseded_by). When a fact becomes DISPUTED or SUPERSEDED, every dependent Assessment and IntelligenceProduct SHALL be flagged review_required with a link to the triggering decision. Dependents are direct (supporting_refs) and indirect (via a supporting Indicator's evidence, a supporting TypologyMatch's indicators, or a supporting Hypothesis's SUPPORTS cells), drafts and finalized alike; the judgement is not changed, a flagged draft SHALL NOT be finalized (409 REVIEW_REQUIRED), clearing the flag is a review action, and the dependents list shows only dependents the caller may read. [v0.1.2 · CR-14-11] Published products SHALL NOT be mutated; a correction review task SHALL be created instead. History SHALL be preserved.
Rationale	Ensures corrections propagate to dependent analysis without rewriting history.
Verification	Pilot: record a source claim; create a PROVISIONAL fact supported by it; establish it; dispute it with contradicting evidence; supersede it. Verify dependent assessment/product are flagged, the published product is unchanged, a correction review task exists, and full history is retrievable.
Priority	MVP / P0
Traceability	F-EVD-008; ASM-01; DIS-01

SRS-FR-DOC-001 — Document ingestion

Requirement	The system SHALL ingest PDF, DOCX, XLSX, CSV, TXT, and common image formats while preserving original bytes.
Rationale	Supports core evidence intake.
Verification	Upload representative formats and verify originals plus metadata.
Priority	MVP / P0
Traceability	F-DOC-001

SRS-FR-DOC-002 — OCR and text extraction [v0.1.4 · CR-N-09]

Requirement	The system SHALL extract searchable text from evidence items (OCR of page images, PDF text layers, plain text) as a page-mapped, review-required DerivedText with engine, version, models and per-page confidence, without modifying the original. Machine output SHALL remain REVIEW_REQUIRED until a person records a review. The feature SHALL be disableable per deployment; automatic extraction after ingest SHALL be off by default.
Rationale	Makes scanned evidence searchable and citable without mistaking machine text for the original.
Verification	Extract a scanned PDF; verify the original row, bytes and version are unchanged, pages map to the input, the record is REVIEW_REQUIRED and a review records reviewer and time.
Priority	P1
Traceability	F-DOC-002; EVD-02; TEC-01

SRS-FR-DOC-003 — Machine translation [v0.1.4 · CR-N-10]

Requirement	The system SHALL translate evidence text (Indonesian ↔ English in v0.1.4) only with an offline engine, keep the source-language text of every page, label the output "Machine translation — review required" until a person reviews it, never translate a translation, and carry the machine-translation label into any intelligence product that uses it.
Rationale	Daily translation need without sending evidence content to a third party.
Verification	Translate with network access disabled; verify source text, label, lineage (translation → OCR derivative → original) and 409 on translating a translation.
Priority	P1
Traceability	EVD-02; TEC-01; PRI-01

6.3 Entity, Relationship and Asset

SRS-FR-ENT-001 — Entity registry

Requirement	The system SHALL support canonical Entity records for person, organisation/company, account, wallet, address, domain, phone/email, government agency, and extensible types.
Rationale	Provides reusable identity layer.
Verification	Create each baseline type and verify unique stable identifiers.
Priority	MVP / P0
Traceability	F-ENT-001; ENT-01

SRS-FR-ENT-002 — Aliases and identifiers

Requirement	The system SHALL allow conflicting aliases/identifiers to coexist with separate provenance and status.
Rationale	Preserves uncertainty and contradictory data.
Verification	Add conflicting identifiers and verify neither overwrites the other.
Priority	MVP / P0
Traceability	F-ENT-002

SRS-FR-ENT-003 — Candidate matching

Requirement	The system SHALL present matching and conflicting features for possible duplicates before merge.
Rationale	Reduces false entity consolidation.
Verification	Create similar entities and verify comparison view.
Priority	MVP / P0
Traceability	F-ENT-003; ENT-01

SRS-FR-ENT-004 — Merge and unmerge

Requirement	The system SHALL support evidence-based merge with rationale and SHALL support unmerge that restores prior records and relationships without erasing history. Each merge and unmerge SHALL be recorded as a ResolutionDecision (MERGE / UNMERGE, the latter referencing the reversed MERGE) per SRS-FR-ENT-006. <i>[v0.1.1 · ER]</i> Every MERGE and UNMERGE is high-impact in the MVP: it SHALL name a reviewer holding REVIEWER or LEAD membership on every subject entity's case (422 otherwise), and a reviewer who is also the decider SHALL be rejected with 409 STATE_CONFLICT. <i>[v0.1.2 · CR-I2-10, CR-I2-09]</i> Merge is logical repointing through canonical_parent: relationships, claims and absorbed records are never rewritten, so unmerge removes only what the merge added to the survivor and restores its prior handling. <i>[v0.1.2 · CR-I2-13]</i>
Rationale	Protects analytical reversibility.
Verification	Merge then unmerge test entities; compare pre/post state and audit log.
Priority	MVP / P0
Traceability	F-ENT-004; AUD-01

SRS-FR-ENT-005 — Entity resolution status

Requirement	The system SHALL support the entity resolution states UNRESOLVED, RESOLVED, CONFLICTED, MERGED, SPLIT (Data Model §8.1) and SHALL expose status in search/graph views. Candidate and probable matches SHALL be shown as POSSIBLE_MATCH ResolutionDecisions, not as entity states. v0.1 terms map as: confirmed → RESOLVED; disputed → CONFLICTED; unresolved → UNRESOLVED; candidate/probable → POSSIBLE_MATCH decision. <i>[v0.1.1 · ER]</i>
Rationale	Keeps identity uncertainty visible.
Verification	Record resolution decisions that change status and verify graph/search labels. <i>[v0.1.1 · ER]</i>
Priority	MVP / P0
Traceability	F-ENT-005

SRS-FR-ENT-006 — Resolution decision record *[v0.1.1 · ER]*

Requirement	The system SHALL record every merge, unmerge, keep-separate, possible-match and defer decision as an append-only ResolutionDecision with evidence, confidence, rationale and reviewer where required, and SHALL derive entity resolution state only from these decisions.
Rationale	Separates decisions about pairs/sets of records from entity state, so identity judgements are reviewable, reversible and auditable. Approved by product owner, 2026-10-08.
Verification	Record POSSIBLE_MATCH, KEEP_SEPARATE, DEFER, MERGE and UNMERGE decisions on test entities; verify each is retrievable in order, cannot be edited or deleted, and that resolution_status changes only as their effect; attempt a direct status update and a MERGE without surviving_entity_ref and verify rejection; verify every MERGE/UNMERGE requires a reviewer other than the decider (missing → 422, reviewer = decider → 409) <i>[v0.1.2 · CR-I2-10, CR-I2-09]</i> ; verify a KEEP_SEPARATE pair is not re-suggested until new evidence is attached.
Priority	MVP / P0

Requirement	The system SHALL record every merge, unmerge, keep-separate, possible-match and defer decision as an append-only ResolutionDecision with evidence, confidence, rationale and reviewer where required, and SHALL derive entity resolution state only from these decisions.
Traceability	F-ENT-003; F-ENT-004; F-ENT-005; ENT-01; AUD-01

SRS-FR-REL-001 — First-class relationship

Requirement	The system SHALL represent material relationships as canonical objects with type, endpoints, evidence, dates, confidence, status, and analyst.
Rationale	Prevents graph-only unsupported edges.
Verification	Create relationship and trace from graph to evidence.
Priority	MVP / P0
Traceability	F-REL-001; REL-01

SRS-FR-REL-002 — Ownership interest

Requirement	The system SHALL represent direct/indirect ownership with percentage or unknown, dates, and evidence. A BENEFICIAL, ECONOMIC_INTEREST or NOMINEE_ASSERTED interest SHALL become ESTABLISHED only when an ESTABLISHED Fact is among its supporting evidence (422 otherwise). <i>[v0.1.2 · CR-I3-11]</i>
Rationale	Supports beneficial-ownership analysis without fabricated zeroes.
Verification	Create unknown and numeric ownership records; verify semantics.
Priority	MVP / P0
Traceability	F-REL-002

SRS-FR-REL-003 — Control distinction

Requirement	The system SHALL distinguish legal ownership, beneficial ownership, control, use, and association.
Rationale	Avoids collapsing materially different relationships.
Verification	Create each relation type and verify distinct UI/API representation.
Priority	MVP / P0
Traceability	F-REL-003; AST-01

SRS-FR-AST-001 — Asset registry

Requirement	The system SHALL support property, vehicle, shares, vessel, aircraft, crypto and extensible asset types with attribution type and evidence.
Rationale	Supports asset tracing.
Verification	Create asset with ownership and control assertions.
Priority	MVP / P0
Traceability	F-AST-001

6.4 Event, Timeline and Value Flow

SRS-FR-TIM-001 — Event records

Requirement	The system SHALL store events with type, date/date-range, date precision, linked entities/assets, location where relevant, and evidence. Date precision uses the registry enum <code>temporal_precision</code> (DATETIME, DAY, MONTH, YEAR, RANGE, UNKNOWN) and the date string shape SHALL match it; an unknown date is recorded as <code>start_time: null</code> with precision UNKNOWN and is permitted only with that precision. <i>[v0.1.2 · CR-I3-06]</i>
Rationale	Supports temporal analysis.
Verification	Create exact/approximate events and verify precision retained.
Priority	MVP / P0
Traceability	F-TIM-001

SRS-FR-TIM-002 — Visual timeline

Requirement	The system SHALL render filtered timelines for case/entity/asset and link each item to canonical event/evidence.
Rationale	Enables chronological reconstruction.
Verification	Filter timeline and open source event.
Priority	MVP / P0
Traceability	F-TIM-002

SRS-FR-VAL-001 — Value-flow record

Requirement	The system SHALL require origin, destination, mechanism, value/range/unknown, currency where applicable, date/date-range, evidence, confidence, and mandatory <code>flow_class</code> . Flow, leg and event evidence MAY reference <code>EvidenceExtract</code> , <code>EvidenceItem</code> or <code>Fact</code> ; <code>DIRECT</code> and <code>DOCUMENTED</code> flows and legs SHALL reference at least one <code>EvidenceItem</code> or <code>EvidenceExtract</code> (a <code>Fact</code> alone → 422). <i>[v0.1.2 · CR-I3-10]</i> A class change that raises certainty — order used only for this invariant: <code>HYPOTHETICAL < RECONSTRUCTED < DOCUMENTED < DIRECT</code> — SHALL add at least one evidence reference not previously attached; a flow SHALL never be stronger than its weakest leg; lowering a class SHALL meet the target class's requirements. <i>[v0.1.2 · CR-I3-09]</i>
Rationale	Makes value movement explicit and qualified.
Verification	Create all four flow classes and verify mandatory fields.
Priority	MVP / P0
Traceability	F-VAL-001; VAL-01

SRS-FR-VAL-002 — Multi-leg flows

Requirement	The system SHALL support multi-leg value-flow chains in which every leg retains separate evidence, class, value, and confidence. Each leg carries its own confidence and, for <code>RECONSTRUCTED/HYPOTHETICAL</code> legs, a <code>reconstruction_basis</code> . Legs are immutable and append-only: appended in order (<code>sequence = n + 1</code>), each starting where the previous one ended; a complete chain takes no more legs (409) and the endpoints of a chained flow are fixed (409). <i>[v0.1.2 · CR-I3-04]</i>
Rationale	Avoids certainty inheritance across a chain.
Verification	Create 3-leg mixed-class flow and inspect each leg.
Priority	MVP / P0
Traceability	F-VAL-002

SRS-FR-VAL-003 — Flow visualisation

Requirement	The system SHALL visually distinguish DIRECT, DOCUMENTED, RECONSTRUCTED, and HYPOTHETICAL flows in interactive and exported views.
Rationale	Prevents inference from looking like direct transaction evidence.
Verification	Export mixed-class flow and verify class distinction persists.
Priority	MVP / P0
Traceability	F-VAL-003

SRS-FR-VAL-004 — Unknown and ranges

Requirement	The system SHALL represent unknown, minimum, maximum, and approximate values without coercing unknown to numeric zero. Amount precision uses the registry enum <code>money_precision</code> (EXACT, APPROXIMATE, ESTIMATED); a range is never EXACT. <i>[v0.1.2 · CR-I3-06]</i> Aggregates SHALL be computed per (flow_class, flow_type, currency) with no grand total; CONTRACT/SUBCONTRACT groups are obligations, only DIRECT groups are settlement-evidenced; lower/upper/exact totals count unknown amounts and never treat them as 0; legs are never added to their flow's totals. <i>[v0.1.2 · CR-I3-12]</i>
Rationale	Preserves uncertainty.
Verification	Store unknown/range values and inspect API/UI.
Priority	MVP / P0
Traceability	F-VAL-004

6.5 Typology, Hypothesis and Assessment

SRS-FR-TYP-001 — Typology catalogue

Requirement	The system SHALL expose versioned CS-AML typology entries with definition, mechanism, observables, indicators, counter-indicators, false positives, and references. Each catalogue indicator carries its A13 source-mapping origin and verification_status; the catalogue is read-only reference data readable by every signed-in principal (GET /typology-catalogue, /typologies, /typologies/{typologyId}). <i>[v0.1.2 · CR-I4-01]</i>
Rationale	Supports consistent analysis.
Verification	Open typology and verify version and fields.
Priority	MVP / P0
Traceability	F-TYP-001; TYP-01

SRS-FR-TYP-002 — Indicator capture

Requirement	The system SHALL allow indicators and counter-indicators to link to evidence and analytical objects. A case indicator either references a catalogue indicator (class taken from the catalogue; a contradicting class → 422) or uses a local LOCAL-... code with an explicit class; it needs ≥1 evidence reference (EvidenceExtract, EvidenceItem or Fact) and ≥1 subject (Entity, Asset, Event, ValueFlow or Relationship). indicator_status transitions: OBSERVED → CORROBORATED / DISPUTED / RETIRED; DISPUTED → OBSERVED / CORROBORATED / RETIRED; RETIRED is terminal; every change needs a rationale; indicators are never deleted and never presented as proof. <i>[v0.1.2 · CR-I4-02]</i>
Rationale	Prevents free-floating allegations.
Verification	Create indicator with evidence and verify assessment traceability.
Priority	MVP / P0
Traceability	F-TYP-002

SRS-FR-TYP-003 — Typology match worksheet

Requirement	The system SHALL support controlled consistency levels and SHALL prevent a single weak indicator from automatically creating a strong typology match. The analyst assigns consistency_level; the server computes a ceiling (Typology Catalogue §4 MINIMUM RULE) and SHALL reject a level above it (422), never raising a level itself. The ceiling thresholds are adopted provisionally and require AML-specialist review. [v0.1.2 · CR-I4-03]
Rationale	Controls overstatement.
Verification	Create single weak indicator and verify no automatic strong match.
Priority	MVP / P0
Traceability	F-TYP-003

SRS-FR-HYP-001 — Competing hypotheses

Requirement	The system SHALL support at least two competing hypotheses per material investigation where alternatives are plausible. The rule is enforced at assessment finalization: finalizing an assessment whose scope_refs contain a hypothesis without a competing hypothesis SHALL be rejected with 409 STATE_CONFLICT, details.reason = "COMPETING_HYPOTHESES_REQUIRED". A hypothesis status change needs a rationale; SUPPORTED needs a SUPPORTS cell and WEAKENED/REJECTED a CONTRADICTS cell (409 otherwise). [v0.1.2 · CR-I4-08] Each hypothesis MAY carry a role (registry enum hypothesis_role: PRINCIPAL, ALTERNATIVE_LEGITIMATE, ALTERNATIVE_MECHANISM, INSUFFICIENT_INFORMATION) and stated assumptions. [v0.1.2 · CR-I4-06]
Rationale	Reduces confirmation bias.
Verification	Create H1/H2 and link supporting/contradicting evidence.
Priority	MVP / P0
Traceability	F-HYP-001; HYP-01

SRS-FR-HYP-002 — Support and contradiction matrix

Requirement	The system SHALL store evidence/indicator effect on each hypothesis as support, contradiction, neutral, or unknown with analyst rationale. Effects use the registry enum hypothesis_link_effect (SUPPORTS, CONTRADICTS, NEUTRAL, UNKNOWN); each cell is an append-only HypothesisLink version, and supporting_refs/contradicting_refs are derived from the current cells. [v0.1.2 · CR-I4-07, CR-I4-04]
Rationale	Makes reasoning inspectable.
Verification	Populate matrix and verify every cell retains rationale/history.
Priority	MVP / P0
Traceability	F-HYP-002

SRS-FR-HYP-003 — Intelligence gaps

Requirement	The system SHALL maintain explicit intelligence gaps capable of affecting an assessment. Gaps are never deleted; closing statuses (PARTIALLY_RESOLVED, RESOLVED, ACCEPTED) need a closure rationale and every status change is kept in the gap's history. [v0.1.2 · CR-I4-05]
Rationale	Keeps unknowns visible.
Verification	Create gap and include it in final assessment.
Priority	MVP / P0
Traceability	F-HYP-003; GAP-01

SRS-FR-ASM-001 — Assessment

Requirement	The system SHALL store judgement, confidence, basis, alternatives, gaps, author, reviewer, and version. Version history is an append-only revision snapshot per record_version (GET /assessments/{assessmentId}/revisions). [v0.1.2 · CR-14-05] Finalized and reviewed are distinct: finalization freezes the content and moves the envelope status DRAFT → FINALIZED while review_status stays DRAFT until a review records PEER_REVIEWED / APPROVED / SUPERSEDED, after which the envelope status follows review_status. [v0.1.2 · CR-14-09]
Rationale	Creates reproducible analytical product.
Verification	Create and revise assessment; verify version history.
Priority	MVP / P0
Traceability	F-ASM-001; ASM-01

SRS-FR-ASM-002 — Confidence model

Requirement	The system SHALL support the confidence levels HIGH, MODERATE, LOW, and INSUFFICIENT_BASIS (wire values from the Data Model Annex A registry; display labels High, Moderate, Low, Insufficient Basis are separate and translatable), each with mandatory rationale. INSUFFICIENT_BASIS means a judgement was attempted but the evidential basis is insufficient; it is not a level below LOW and SHALL NOT be converted to LOW, null, zero, or omitted. Null is permitted only on drafts where no confidence judgement has been made; a finalized assessment SHALL carry a non-null level. No normalization, import, or export SHALL raise certainty. [v0.1.1 · A09]
Rationale	Avoids pseudo-precision.
Verification	Save each level and verify mandatory rationale; round-trip every level through DB/API/UI/export and verify INSUFFICIENT_BASIS is never converted to LOW, null, or zero; attempt to finalize an assessment with a null level and verify rejection. [v0.1.1 · A09]
Priority	MVP / P0
Traceability	F-ASM-002

SRS-FR-ASM-003 — Backward traceability

Requirement	The system SHALL allow a reviewer to traverse an assessment backward through hypotheses/indicators/facts/evidence/sources. The provenance response lists references per kind (including typology matches, sources and gaps) plus the full trace (root, nodes, edges, source paths); scope_refs MAY target Hypothesis, TypologyMatch, Indicator, Entity, Asset, Event, ValueFlow or Relationship, and supporting_refs Fact, Indicator, TypologyMatch, Hypothesis, EvidenceExtract or EvidenceItem (a Source → 422). [v0.1.2 · CR-14-14]
Rationale	Core reproducibility requirement.
Verification	Select key assessment and trace to origin without external notes.
Priority	MVP / P0
Traceability	F-ASM-001; ASM-01 [v0.1.1 · A05]

SRS-FR-ASM-004 — Disconfirming search record [v0.1.1 · A05]

Requirement	The system SHALL require at least one recorded disconfirming-search entry (what was searched, sources consulted, result, rationale; Data Model §13.3 Assessment.disconfirming_searches[]) before a high-impact or adverse assessment (high_impact_adverse), or a product that depends on it, can pass review. The rule SHALL be enforced at review approval, not at finalization: an approval without such an entry SHALL be rejected with 409 STATE_CONFLICT and details.reason = "DISCONFIRMATION_REQUIRED". Entries are recorded through POST /assessments/{assessmentId}/disconfirming-searches. [v0.1.1 · C10] Entries MAY be appended to a DRAFT or FINALIZED assessment until the assessment (or a product depending on it) passes review approval; afterwards the record is frozen (409 STATE_CONFLICT). Finalization does not freeze the disconfirming-search record. [v0.1.2 · CR-I4-10]
Rationale	Ensures evidence that could weaken adverse findings has been sought and recorded; tested separately from backward traceability (SRS-FR-ASM-003).
Verification	Attempt to approve the review of a high-impact adverse assessment without a disconfirming-search entry; verify 409 STATE_CONFLICT with details.reason = "DISCONFIRMATION_REQUIRED". Record an entry; verify the review approval succeeds. [v0.1.1 · C10] Finalize an assessment, append a disconfirming search and verify it is accepted; approve the review and verify a further append returns 409. [v0.1.2 · CR-I4-10]
Priority	MVP / P0
Traceability	F-ASM-003; HYP-02; QUA-01 [v0.1.1 · A05]

6.6 Search and Graph

SRS-FR-SCH-001 — Full-text and object search

Requirement	The system SHALL search authorised cases, entities, sources, evidence metadata, extracts, assets, events, and products using full-text and structured filters. MVP 0.1 implements it as GET /search (API §17). Indexed: cases, entities (names, aliases, identifiers), sources, evidence metadata, extracts (cited text), claims, facts, relationships (type and dates only, never endpoint names), assets, events, value flows and products. Evidence file content (OCR / full text) is not indexed in MVP 0.1 (Phase 2); hypotheses, assessments and indicators are not indexed yet. Every hit links to its canonical object. [v0.1.3 · CR-I5-01, CR-I5-07] Since v0.1.4 completed DerivedText is indexed as DERIVED_TEXT with the epistemic marker DERIVED (never presented as the original) and can be switched off per deployment. [v0.1.4 · CR-N-11]
Rationale	Supports investigation retrieval.
Verification	Search known fixture and verify expected scoped results.
Priority	MVP / P0
Traceability	F-SCH-001

SRS-FR-SCH-002 — Permission-filtered results

Requirement	The system SHALL apply object-level authorization before returning search results or counts. Authorization SHALL also precede facets, snippets and ordering. SOURCE_PROTECTED material (and objects whose only readable context is a SOURCE_PROTECTED case) SHALL be returned only on explicit request (include_source_protected=true) by a principal holding the per-case protected-source grant; otherwise it SHALL produce no hit, count, facet or snippet. [v0.1.3 · CR-I5-04] Rate limits: search 120 and graph 60 requests per minute per user; search statement timeout 5 s. The residual timing channel (query latency depends on the total corpus size) is accepted for MVP 0.1 with synthetic data only and SHALL be re-reviewed before real (non-synthetic) data is processed. [v0.1.3 · CR-I5-08]
Rationale	Prevents existence leakage.

Requirement	The system SHALL apply object-level authorization before returning search results or counts. Authorization SHALL also precede facets, snippets and ordering. SOURCE_PROTECTED material (and objects whose only readable context is a SOURCE_PROTECTED case) SHALL be returned only on explicit request (include_source_protected=true) by a principal holding the per-case protected-source grant; otherwise it SHALL produce no hit, count, facet or snippet. [v0.1.3 · CR-I5-04] Rate limits: search 120 and graph 60 requests per minute per user; search statement timeout 5 s. The residual timing channel (query latency depends on the total corpus size) is accepted for MVP 0.1 with synthetic data only and SHALL be re-reviewed before real (non-synthetic) data is processed. [v0.1.3 · CR-I5-08]
Verification	Compare user with/without access and verify hidden objects do not affect result counts. Repeat with and without the protected-source grant and with and without include_source_protected; verify that counts and facets change only for the authorized, opted-in principal. [v0.1.3 · CR-I5-04]
Priority	MVP / P0
Traceability	F-SCH-002; SEC-01

SRS-FR-GRF-001 — Graph exploration

Requirement	The system SHALL display authorised entities and canonical relationships with evidence/status/confidence access from each edge. The MVP graph is the per-case, policy-filtered projection GET /cases/{caseId}/graph (entities, assets and events as nodes; relationships with ownership/control details, value flows and legs with their own class, and event participation as edges), computed on demand and marked derived. [v0.1.2 · CR-I3-07] Cross-object exploration uses POST /graph/query and POST /graph/paths (API §18): depth ≤ 3, 500 nodes, 1,500 edges and a 4 s wall-clock budget return partial results with stated truncation reasons; only the 5 s statement timeout is an error (503). Shared-attribute leads (same normalized identifier, phone, e-mail or address) are labelled candidates and SHALL NOT be stored, drawn as edges or treated as facts or merges. [v0.1.3 · CR-I5-02, CR-I5-05, CR-I5-06]
Rationale	Supports network analysis without hiding provenance.
Verification	Expand graph and open edge evidence.
Priority	MVP / P0
Traceability	F-GRF-001

6.7 Intelligence Product, Review and Dissemination

SRS-FR-PRD-001 — Product templates

Requirement	The system SHALL generate Financial Intelligence Note, Entity Profile, Asset Profile, Network Analysis, Referral Package, and Case Report from canonical objects. Each template defines typed sections (SUBJECTS, FACTS, CLAIMS, ANALYSIS, UNKNOWN, CONTEXT) bound to canonical objects; submission freezes a version with its rendered document and content hash. [v0.1.3 · CR-I6-01]
Rationale	Standardises outputs.
Verification	Generate each baseline template with metadata/evidence index.
Priority	MVP / P0
Traceability	F-PRD-001

SRS-FR-REV-001 — Peer review workflow

Requirement	The system SHALL support comment, request-change, approve, and reject; high-impact approval SHALL be attributable and independent where policy requires. Review kinds: product version, assessment, review_required clearance, handling change and correction. Authors, contributors and the requester SHALL NOT decide a review. [v0.1.3 · CR-I6-02, CR-I6-03]
Rationale	Implements quality gate.
Verification	Author attempts self-approval under independent-review policy and is denied.
Priority	MVP / P0
Traceability	F-REV-001; QUA-01

SRS-FR-PRD-002 — Versioning and corrections

Requirement	The system SHALL preserve prior intelligence-product versions and mark superseded/ corrected versions explicitly. A correction opens version n+1; earlier versions remain valid until the correction is approved, then become SUPERSEDED with their disseminations. Withdrawal retracts every version and revokes disseminations. [v0.1.3 · CR-I6-05]
Rationale	Prevents historical rewriting.
Verification	Issue correction and verify old version remains immutable/readable.
Priority	MVP / P0
Traceability	F-PRD-002; AUD-01

SRS-FR-PRD-003 — Evidence index

Requirement	The system SHALL generate an evidence/source index sufficient to trace every material fact/findings used in a product.
Rationale	Supports review and referral.
Verification	Generate product and click/resolve each material citation.
Priority	MVP / P0
Traceability	F-PRD-003

SRS-FR-DIS-001 — Dissemination approval

Requirement	The system SHALL block external export until required handling classification, recipient, purpose, and approval are recorded. The handling classification of a release SHALL be at least the product's classification; the approver SHALL be a REVIEWER or LEAD case member other than the requester. [v0.1.3 · CR-I6-06]
Rationale	Controls harm and confidentiality.
Verification	Attempt export before approval and verify denial.
Priority	MVP / P0
Traceability	F-DIS-001; DIS-01

SRS-FR-DIS-002 — Secure export

Requirement	The system SHALL export only explicitly approved objects and SHALL support minimisation/redaction. SOURCE_PROTECTED objects SHALL be exported only when the approval explicitly releases them (approver with the per-case protected-source grant, with rationale); otherwise they, their index entries and citations are withheld and only their count is recorded. [v0.1.3 · CR-I6-07]
Rationale	Prevents accidental over-disclosure.
Verification	Export package and verify excluded restricted object is absent.

Requirement	The system SHALL export only explicitly approved objects and SHALL support minimisation/redaction. SOURCE_PROTECTED objects SHALL be exported only when the approval explicitly releases them (approver with the per-case protected-source grant, with rationale); otherwise they, their index entries and citations are withheld and only their count is recorded. <i>[v0.1.3 · CR-I6-07]</i>
Priority	MVP / P0
Traceability	F-DIS-002; PRI-01

SRS-FR-DIS-003 — Referral package

Requirement	The system SHALL generate structured referral containing subjects, key facts, evidence references, timeline, value flows, indicators, gaps, confidence, and contact point.
Rationale	Makes outputs actionable.
Verification	Generate referral and verify fact/analysis/unknown sections remain distinct.
Priority	MVP / P0
Traceability	F-DIS-003

SRS-FR-DIS-004 — Sharing log

Requirement	The system SHALL record recipient, product/version, date, purpose, restrictions, and approving user for every external dissemination.
Rationale	Maintains accountability.
Verification	Share export and verify immutable log entry.
Priority	MVP / P0
Traceability	F-DIS-004; AUD-01

6.8 Administration, Security and Operations

SRS-FR-ADM-001 — Controlled vocabularies

Requirement	The system SHALL version relationship types, asset types, statuses, classifications, and other controlled vocabularies.
Rationale	Preserves historical semantics.
Verification	Change label and verify historical record retains version context.
Priority	MVP / P0
Traceability	F-ADM-001

SRS-FR-ADM-003 — Retention policies

Requirement	The system SHALL support retention/disposition policy by object type/classification and SHALL audit disposition actions. MVP 0.1: rules apply to evidence items, export packages and cases, optionally narrowed by classification; evaluation reports or proposes, and every disposition needs an approved, logged decision and is refused under an active legal hold. Supported dispositions are DELETE (stored bytes purged, record kept as tombstone), ARCHIVE and REVIEW. ANONYMIZE and whole-case DELETE are refused and deferred to v0.2. <i>[v0.1.3 · CR-I7-01, CR-I7-02, CR-I7-05]</i>
Rationale	Implements data minimisation lifecycle.
Verification	Preview and execute test disposition with audit evidence.
Priority	MVP / P0
Traceability	F-ADM-003; PRI-02

SRS-FR-SEC-001 — OIDC and MFA

Requirement	The system SHALL authenticate via OIDC-compatible identity provider and enforce MFA according to deployment policy.
Rationale	Strong identity control.
Verification	Disable IdP user and verify access revoked; validate MFA path. Verify a high-risk action without a step-up answers 403 STEP_UP_REQUIRED and succeeds after GET /auth/login?acr=2 (SRS-SEC-005). <i>[v0.1.4 · CR-N-13]</i>
Priority	MVP / P0
Traceability	F-SEC-001

SRS-FR-SEC-002 — Role/object access

Requirement	The system SHALL enforce role, case membership, classification, and need-to-know at the API/service layer. Case memberships (roles LEAD, ANALYST, REVIEWER — registry enum case_membership_role) SHALL be manageable through the API (GET/POST /cases/{caseId}/memberships, PATCH/DELETE /case-memberships/{membershipId} with If-Match) by the case LEAD; revocation is soft and audited. Objects created outside a case path SHALL name 1..20 case_links on creation, each requiring case.update (422/403/404 otherwise). <i>[v0.1.2 · CR-I1-02, CR-I2-01, CR-I3-13, CR-I4-12]</i>
Rationale	Core confidentiality requirement.
Verification	Direct API access to unauthorised object must return denial regardless of UI.
Priority	MVP / P0
Traceability	F-SEC-002; SEC-01

SRS-FR-SEC-003 — Protected-source compartment

Requirement	The system SHALL store protected-source identity under separate authorization so analysts may use source-derived evidence without automatically seeing identity. Access to SOURCE_PROTECTED material requires both the IdP eligibility role csaml-protected-source and the per-case membership grant protected_source_authorized (Technical Stack §10). <i>[v0.1.2 · CR-I1-10]</i>
Rationale	Protects vulnerable sources.
Verification	Analyst can access derived evidence but not source identity.
Priority	MVP / P0
Traceability	F-SEC-003; SEC-02

SRS-FR-AUD-001 — Immutable audit trail

Requirement	The system SHALL record material create/update/delete/merge/unmerge/review/export/access-policy actions and ordinary users SHALL NOT modify audit records.
Rationale	Supports accountability.
Verification	Attempt audit modification as ordinary user and verify denial.
Priority	MVP / P0
Traceability	F-AUD-001

SRS-FR-OPS-001 — Backup and restore

Requirement	The system SHALL back up canonical DB, evidence, configuration, and required keys/metadata using documented recovery procedures and SHALL support periodic restore tests.
Rationale	Provides recoverability.
Verification	Perform restore drill into isolated environment and document result.
Priority	MVP / P0
Traceability	F-OPS-001

7. Data Requirements

SRS-DR-001 — Canonical object set

Requirement	The system SHALL implement at minimum Case, Source, EvidenceItem, EvidenceExtract, Claim, VerificationDecision, Fact, Entity, Relationship, Asset, Event, ValueFlow, Indicator, TypologyMatch, Hypothesis, IntelligenceGap, Assessment, IntelligenceProduct, Review, Dissemination, and AuditEvent as durable domain objects. <i>[v0.1.1 · A10]</i>
Rationale	Aligns SRS with canonical data model.
Verification	Schema review and CRUD contract tests for all required object types.
Priority	MVP
Traceability	Data Model v0.1.4

SRS-DR-002 — Stable identifiers

Requirement	Every canonical object SHALL have a stable, non-recycled identifier independent of display name.
Rationale	Supports audit and linking.
Verification	Rename object and verify identifier unchanged.
Priority	MVP
Traceability	Data Model v0.1.4

SRS-DR-003 — Version semantics

Requirement	Material assessments, charters, products, vocabularies, and other configured material objects SHALL preserve version history.
Rationale	Prevents historical overwrite.
Verification	Update object and retrieve prior version.
Priority	MVP
Traceability	AUD-01

SRS-DR-004 — Temporal semantics

Requirement	Objects with temporal meaning SHALL support valid-time fields and precision/unknown semantics rather than forcing exact dates.
Rationale	Supports imperfect open-source data.
Verification	Store year-only and approximate event date.
Priority	MVP
Traceability	Data Model v0.1.4

SRS-DR-005 — Provenance references

Requirement	Material facts, relationships, asset attributions, value-flow legs, indicators, and assessments SHALL reference supporting evidence and/or source objects.
Rationale	Core evidentiary traceability.
Verification	Run orphan-proposition integrity check.
Priority	MVP
Traceability	SRC-01; REL-01; VAL-01

SRS-DR-006 — Deletion semantics

Requirement	Deletion of canonical investigative objects SHALL be soft-delete/tombstone or equivalent where audit/history obligations require preservation; hard deletion SHALL be controlled by retention/disposition policy. In MVP 0.1 hard deletion is limited to the stored bytes of evidence originals and export packages through an approved DELETE disposition; the database record remains as the tombstone and disposed content answers 409. <i>[v0.1.3 · CR-17-05, CR-17-07]</i>
Rationale	Balances history with minimisation.
Verification	Delete test object and verify policy-compliant behaviour.
Priority	MVP
Traceability	PRI-02; AUD-01

8. External Interface Requirements**SRS-IF-001 — Web user interface**

Requirement	The MVP SHALL provide a responsive web interface for supported desktop browsers; mobile support MAY be read-only or limited.
Rationale	Primary analyst workstation assumption.
Verification	Browser compatibility and viewport tests.
Priority	MVP
Traceability	PRD UX

SRS-IF-002 — REST/HTTP API

Requirement	The system SHALL expose versioned authenticated APIs for canonical objects and SHALL enforce identical authorization rules to the UI. Mutations of versioned resources SHALL use If-Match preconditions (missing → 428 PRECONDITION_REQUIRED; stale → 412 PRECONDITION_FAILED; no silent overwrite), except multi-entity commands (merge, unmerge, POST /resolution-decisions, match-candidate decisions), which SHALL carry a body map expected_versions (missing or incomplete → 428; mismatch → 412 with details.current_record_versions) <i>[v0.1.1 · C03]</i> ; 409 STATE_CONFLICT is reserved for workflow/business-state conflicts. Material commands (evidence ingest finalization, merge/unmerge, review/dissemination approval, export package generation) SHALL require an Idempotency-Key. The OpenAPI 3.1 contract <code>contracts/openapi.yaml</code> (PO vertical slice, contract-first) SHALL be conformed to by the implementation and exercised by contract tests; detailed semantics are in API Specification v0.1.4 §28. <i>[v0.1.1 · A04, A11] [v0.1.1 · C12]</i>
Rationale	Enables integration and testability.
Verification	API contract tests plus authorization parity tests.
Priority	MVP
Traceability	Technology Architecture

SRS-IF-003 — OIDC identity provider

Requirement	The system SHALL integrate with an OIDC-compatible provider for authentication and group/claim mapping. Browser authentication SHALL use a server-side session (BFF): the backend is a confidential OIDC client (Authorization Code + PKCE), the browser holds only an HttpOnly, Secure session cookie, tokens are never exposed to JavaScript, and unsafe methods carry a CSRF token. <i>[v0.1.1 · A11]</i>
Rationale	Avoids custom password storage where possible.
Verification	IdP login/logout/disable/claim tests.
Priority	MVP
Traceability	SEC-01

SRS-IF-004 — Object/evidence storage

Requirement	The system SHALL support encrypted object/file storage for evidence and derivatives with immutable-original semantics.
Rationale	Separates binary evidence from transactional records.
Verification	Upload/download/hash/permission tests.
Priority	MVP
Traceability	EVD-01

SRS-IF-005 — Export formats

Requirement	The system SHALL support PDF and machine-readable JSON/CSV where relevant for approved exports; exported analytical semantics SHALL preserve uncertainty/status/classification.
Rationale	Supports human and system recipients.
Verification	Round-trip test representative objects and inspect uncertainty fields.
Priority	MVP
Traceability	DIS-01

9. Security and Privacy Requirements**SRS-SEC-001 — Least privilege**

Requirement	All authenticated access SHALL default deny and be granted by explicit role/object policy.
Rationale	Sensitive investigations require compartmentalisation.
Verification	Permission matrix tests including negative cases.
Priority	MVP
Traceability	SEC-01

SRS-SEC-002 — Encryption in transit

Requirement	All production HTTP/API traffic SHALL use TLS; plaintext administrative or application access SHALL be disabled except explicitly isolated development environments.
Rationale	Protects credentials and case data.
Verification	TLS configuration scan.
Priority	MVP

Requirement	All production HTTP/API traffic SHALL use TLS; plaintext administrative or application access SHALL be disabled except explicitly isolated development environments.
Traceability	Technology Architecture

SRS-SEC-003 — Encryption at rest

Requirement	Canonical databases, evidence stores, backups, and protected-source data SHALL use encryption at rest appropriate to deployment risk.
Rationale	Protects stolen media/snapshots.
Verification	Configuration and restore tests.
Priority	MVP
Traceability	SEC-02

SRS-SEC-004 — Secrets management

Requirement	Application secrets SHALL NOT be embedded in source code or container images and SHALL be stored/rotated through an approved secret-management mechanism.
Rationale	Reduces credential leakage.
Verification	Repository/image scan and rotation test.
Priority	MVP
Traceability	Technology Architecture

SRS-SEC-005 — Session control

Requirement	Sessions SHALL support idle timeout, absolute lifetime, revocation, and reauthentication for configured high-risk actions. Baseline high-risk actions (reauthentication = ACR 2, password + TOTP, within 900 s by default): dissemination.approve, export.generate, export.download, disposition.approve, disposition.execute, legal_hold.release, entity.merge, entity.unmerge, gate.approve, protected_source.grant, protected_source.access (API §4). Without a recent step-up the action answers 403 STEP_UP_REQUIRED and is audited. A failed sign-in callback SHALL return the browser to the application with a reason code (303), never a raw error. Per-deployment action lists and phishing-resistant factors for approvers are deferred to v0.2. [v0.1.4 · CR-N-13, CR-N-14]
Rationale	Reduces account takeover impact.
Verification	Session expiry/revocation test.
Priority	MVP
Traceability	SEC-01

SRS-SEC-006 — Protected source isolation

Requirement	Protected-source identity SHALL be logically separated from routine case content and SHALL use a narrower permission set.
Rationale	Protects human sources.
Verification	Cross-role authorization test.
Priority	MVP
Traceability	SEC-02

SRS-SEC-007 — Data minimisation

Requirement	The system SHALL support classification, purpose/scope linkage, retention state, and controlled export so sensitive data can be minimised throughout the lifecycle.
Rationale	Implements CSO privacy guardrails.
Verification	Retention/export/minimisation workflow test.
Priority	MVP
Traceability	PRI-01; PRI-02

SRS-SEC-008 — Security logging

Requirement	Security logs SHALL capture authentication, authorization failures, admin changes, and export events without unnecessarily recording evidence payloads or protected-source content.
Rationale	Balances detection and confidentiality.
Verification	Log inspection with synthetic sensitive fields.
Priority	MVP
Traceability	AUD-01

10. Audit and Provenance Requirements

SRS-AUD-001 — Audit event content

Requirement	Material audit events SHALL include actor, action, object ID/type, timestamp, case/context when applicable, and before/after or change reference where appropriate.
Rationale	Allows reconstruction.
Verification	Inspect representative audit records.
Priority	MVP
Traceability	AUD-01

SRS-AUD-002 — Audit integrity

Requirement	Ordinary application users SHALL NOT alter or delete audit events. Administrative maintenance SHALL itself be logged.
Rationale	Prevents tampering.
Verification	Role test and admin-maintenance audit test.
Priority	MVP
Traceability	AUD-01

SRS-AUD-003 — Provenance traversal

Requirement	The system SHALL provide forward/backward traversal between Source, Evidence, derived artefacts, Facts, Indicators, Hypotheses, Assessments and Products.
Rationale	Makes reasoning reviewable.
Verification	Traverse complete pilot-case chain.
Priority	MVP
Traceability	ASM-003; EVD-02

SRS-AUD-004 — Merge history

Requirement	Entity merge/unmerge decisions SHALL record rationale, actor, timestamp, compared features, supporting evidence, and resulting mapping, as ResolutionDecision records (SRS-FR-ENT-006). <i>[v0.1.1 · ER]</i>
Rationale	Critical identity-control history.
Verification	Merge/unmerge audit test.
Priority	MVP
Traceability	ENT-01

11. Performance and Scalability

SRS-NFR-PERF-001 — Interactive latency

Requirement	For a reference MVP deployment, 95% of ordinary authenticated page/API reads SHOULD complete within 2 seconds excluding large file transfer, complex graph traversal, OCR, exports, and external connectors.
Rationale	Usability baseline.
Verification	Load test reference dataset.
Priority	MVP target
Traceability	PRD NFR

SRS-NFR-PERF-002 — Search latency

Requirement	Permission-filtered search SHOULD return first-page results within 3 seconds at reference MVP scale. Both targets apply: global permission-filtered search, first page ≤ 3 s; case-scoped search, p95 < 2 s (Technical Stack §29). They are SHOULD-level targets measured on the reference corpus (SRS-NFR-PERF-004), not guarantees. <i>[v0.1.3 · CR-15-09]</i>
Rationale	Analyst productivity.
Verification	Search benchmark under concurrent users.
Priority	MVP target
Traceability	CAP-10 (Search / Graph / Analytics, product capability registry) <i>[v0.1.1 · A07]</i>

SRS-NFR-PERF-003 — Upload size

Requirement	The deployment SHALL define configurable maximum upload size and SHALL fail safely without partial canonical records when exceeded or interrupted.
Rationale	Evidence reliability.
Verification	Boundary upload tests.
Priority	MVP target
Traceability	F-DOC-001

SRS-NFR-PERF-004 — Reference scale

Requirement	MVP SHALL support at least 25 concurrent authenticated users, 100,000 canonical entities/relationships combined, and 100 GB evidence storage in a reference small-organisation deployment, subject to deployment profile tuning. The reference corpus for search and graph measurements is the synthetic generator of the reference implementation (backend/tests/synthetic_corpus.py, about 97,000 search rows); measurements on it are evidence for the targets, not a guarantee. <i>[v0.1.3 · CR-15-09]</i>
Rationale	Provides engineering sizing target.

Requirement	MVP SHALL support at least 25 concurrent authenticated users, 100,000 canonical entities/relationships combined, and 100 GB evidence storage in a reference small-organisation deployment, subject to deployment profile tuning. The reference corpus for search and graph measurements is the synthetic generator of the reference implementation (backend/tests/synthetic_corpus.py, about 97,000 search rows); measurements on it are evidence for the targets, not a guarantee. [v0.1.3 · CR-15-09]
Verification	Synthetic scale/load test.
Priority	MVP target
Traceability	Technology Architecture

12. Reliability, Backup and Disaster Recovery

SRS-NFR-REL-001 — Transactional consistency

Requirement	Canonical state changes that span related records SHALL use transactional or compensating mechanisms to avoid orphan/inconsistent material records.
Rationale	Protects data integrity.
Verification	Fault-injection transaction tests.
Priority	MVP
Traceability	Data Model

SRS-NFR-REL-002 — Backup coverage

Requirement	Backups SHALL include canonical DB, evidence metadata and files, configuration required for restore, and protected-source data according to classification.
Rationale	Complete recovery.
Verification	Backup inventory review.
Priority	MVP
Traceability	OPS-001

SRS-NFR-REL-003 — Restore testing

Requirement	A restore procedure SHALL be executable into an isolated environment and SHOULD be tested at least quarterly for production deployments.
Rationale	Proves recoverability.
Verification	Documented restore drill.
Priority	MVP
Traceability	OPS-001

SRS-NFR-REL-004 — RPO/RTO profile

Requirement	Deployment documentation SHALL declare target RPO and RTO. MVP reference target SHOULD be RPO ≤24h and RTO ≤8h unless organisational risk requires stricter objectives.
Rationale	Makes resilience measurable.
Verification	DR plan review and timed restore.
Priority	MVP
Traceability	Technology Architecture

13. Search, Graph and Analytical Requirements

SRS-AN-001 — Canonical projection

Requirement	Search and graph stores SHALL be rebuildable projections from canonical objects and SHALL retain canonical object identifiers.
Rationale	Avoids projection becoming truth source.
Verification	Delete/rebuild projection and compare object identity.
Priority	MVP/P1 where noted
Traceability	Technology Architecture

SRS-AN-002 — Uncertainty propagation

Requirement	Graph/search/analytics views SHALL display or expose material status/confidence/class fields used by the canonical records.
Rationale	Prevents visual certainty inflation.
Verification	Inspect disputed entity and reconstructed flow in all views.
Priority	MVP/P1 where noted
Traceability	Technology Architecture

SRS-AN-003 — Path semantics

Requirement	Any path/graph traversal result SHALL be presented as a path over existing relationships, not as proof of a direct relationship between endpoints.
Rationale	Prevents inference misuse.
Verification	Path-result wording/UI test.
Priority	MVP/P1 where noted
Traceability	TEC-02

14. AI and Automation Requirements

SRS-AI-001 — Derived status

Requirement	AI/OCR/NLP-generated outputs SHALL be marked as derived/candidate material with model/tool version and source context when material. OCR, extraction and translation outputs carry automated: true, engine, version and model hashes (Data Model §7.7). <i>[v0.1.4 · CR-N-09, CR-N-10]</i>
Rationale	Prevents automation from becoming evidence.
Verification	Generate candidate and inspect metadata.
Priority	P1 unless AI disabled
Traceability	TEC-01

SRS-AI-002 — Human verification

Requirement	AI output SHALL NOT become a material fact, canonical entity merge, adverse assessment, or dissemination approval without explicit human verification.
Rationale	Preserves accountability.
Verification	Attempt automated promotion and verify blocked state.
Priority	P1 unless AI disabled

Requirement	AI output SHALL NOT become a material fact, canonical entity merge, adverse assessment, or dissemination approval without explicit human verification.
Traceability	TEC-01

SRS-AI-003 — Authorization boundary

Requirement	AI services SHALL receive only content the invoking user/service is authorised to access; retrieval for AI SHALL enforce the same object policies.
Rationale	Prevents cross-case leakage.
Verification	Prompt/AI retrieval authorization tests.
Priority	P1 unless AI disabled
Traceability	SEC-01

SRS-AI-004 — Optional feature flag

Requirement	AI assistance SHALL be disableable at deployment/environment level without breaking core investigative workflows.
Rationale	Supports high-risk/offline environments.
Verification	Disable AI and execute MVP pilot case.
Priority	P1 unless AI disabled
Traceability	F-ADM-004

SRS-AI-005 — Offline-only automation [v0.1.4 · CR-N-10]

Requirement	Every automation engine (OCR, machine translation and any future AI) SHALL run offline on the CS-AML host with pinned, integrity-checked models; evidence content SHALL NOT be sent to a third-party service and there SHALL be no online fallback. Reviewed translations need no second reviewer, but the machine-translation label SHALL be carried into any intelligence product that uses the translation.
Rationale	Privacy and source protection: evidence content never leaves the host.
Verification	Run OCR and translation with outbound network disabled; verify no network library is used and a modified model file is refused.
Priority	MVP where automation is enabled
Traceability	TEC-01; PRI-01; SEC-01

15. UX and Accessibility Requirements

SRS-UX-001 — Evidence-first navigation

Requirement	Material analytical records SHALL expose a direct route to supporting evidence/source in no more than a small number of user actions.
Rationale	Review efficiency.
Verification	Usability test with analyst.
Priority	MVP
Traceability	PRD UX

SRS-UX-002 — Visible uncertainty

Requirement	Candidate/disputed entities, inferred relationships, and reconstructed/hypothetical flows SHALL be visually distinguishable from confirmed/direct records.
Rationale	Core analytical safety.
Verification	Visual regression/usability test.
Priority	MVP
Traceability	Technology Architecture

SRS-UX-003 — Destructive action confirmation

Requirement	Merge, unmerge, disposition, external export, and other high-impact actions SHALL require explicit confirmation and appropriate authorization.
Rationale	Reduces accidental harm.
Verification	Negative/confirmation tests.
Priority	MVP
Traceability	Control Guide

SRS-UX-004 — Accessibility baseline

Requirement	Primary workflows SHOULD meet WCAG 2.2 AA principles where feasible, including keyboard navigation, labels, contrast, and non-colour-only meaning.
Rationale	Inclusive use.
Verification	Automated plus manual accessibility checks.
Priority	MVP
Traceability	Product requirement

16. Administration and Configuration

SRS-ADM-001 — Policy configuration

Requirement	Admins SHALL configure classifications, role mappings, controlled vocabularies, retention rules, and feature flags without direct database editing.
Rationale	Safe administration.
Verification	Admin UI/API configuration tests.
Priority	MVP
Traceability	F-ADM-*

SRS-ADM-002 — Configuration audit

Requirement	Material policy/configuration changes SHALL be versioned or audit-logged with actor/time.
Rationale	Change accountability.
Verification	Change policy and inspect audit.
Priority	MVP
Traceability	AUD-01

SRS-ADM-003 — No silent retroactive rewrite

Requirement	Vocabulary/template changes SHALL NOT silently rewrite historical analytical meaning.
Rationale	Preserves historical context.
Verification	Rename vocabulary and inspect old record semantics.
Priority	MVP
Traceability	F-ADM-001

17. Interoperability and Integration

SRS-INT-001 — Connector provenance

Requirement	Any external connector SHALL retain provider/source, remote identifier, retrieval timestamp, mapping version, and licence/terms metadata where relevant.
Rationale	Makes enrichment attributable.
Verification	Connector fixture import test.
Priority	P1 except architectural rule
Traceability	SRC-01

SRS-INT-002 — No silent merge

Requirement	Imported external entity candidates SHALL NOT silently merge into canonical entities solely because an external identifier or fuzzy match is returned.
Rationale	Prevents external false positives.
Verification	Import ambiguous candidate and verify review required.
Priority	P1 except architectural rule
Traceability	ENT-01

SRS-INT-003 — FollowTheMoney mapping

Requirement	Where implemented, FollowTheMoney import/export SHALL preserve unmapped CS-AML fields and mapping-version metadata.
Rationale	Maintains interoperability without semantic loss.
Verification	Round-trip mapping test.
Priority	P1 except architectural rule
Traceability	F-INT-003

SRS-INT-004 — Projection connectors

Requirement	Graph/search integrations SHALL be rebuildable and SHALL not own unique canonical analytical facts.
Rationale	Architectural consistency.
Verification	Projection rebuild test.
Priority	P1 except architectural rule
Traceability	Technology Architecture

18. Deployment and Operations

SRS-OPS-001 — Containerised/reference deployment

Requirement	A reference deployment SHOULD be reproducible using containerised or equivalent declarative deployment artefacts with environment-specific secrets externalised.
Rationale	Repeatable operations.
Verification	Fresh deployment from documentation.
Priority	MVP
Traceability	Technology Architecture

SRS-OPS-002 — Health endpoints

Requirement	Application services SHALL provide health/readiness indicators that do not reveal case-sensitive content. MVP 0.1: GET /api/v1/health (unauthenticated) reports only status and latency of the database, Valkey and the object store plus build version; the Prometheus /metrics endpoint is internal and never routed externally. [v0.1.3 · CR-17-08]
Rationale	Operational monitoring.
Verification	Health endpoint test.
Priority	MVP
Traceability	OPS

SRS-OPS-003 — Migration safety

Requirement	Schema migrations SHALL be versioned, reversible where practical, and tested against representative backup data before production application.
Rationale	Protects canonical records.
Verification	Migration dry-run/rollback test.
Priority	MVP
Traceability	Data Model

SRS-OPS-004 — Observability minimisation

Requirement	Metrics/logs/traces SHALL avoid storing evidence bodies, protected-source identity, or other case-sensitive payloads unless explicitly required and protected.
Rationale	Prevents telemetry leakage.
Verification	Telemetry inspection.
Priority	MVP
Traceability	SEC-02

19. Verification and Acceptance

Each mandatory requirement SHALL be verifiable by one or more of: automated unit/integration/API test, security test, data-integrity test, UI acceptance test, operational drill, or structured manual review. A requirement is not complete merely because code exists.

Code	Verification method
V1	Automated functional/API test
V2	Authorization/security negative test
V3	Data integrity/schema/provenance test
V4	UI/usability acceptance test
V5	Operational backup/restore/deployment drill
V6	Structured reviewer sign-off

20. Traceability Matrix

SRS family	Upstream product	Framework/control	Canonical data	Primary test
CASE	F-CASE-*	GOV-01/CAS-01/CAS-02	Case, Review, AuditEvent	V1/V6
EVD/CLM	F-EVD-*/F-DOC-001	SRC-01/SRC-02/EVD-01/ EVD-02/QUA-01	Source, EvidenceItem, EvidenceExtract, Claim, VerificationDecision, Fact	V1/V3/V6 [v0.1.1 · A10]
ENT/REL/AST	F-ENT-*/F-REL-*/ F-AST-001	ENT-01/REL-01/AST-01	Entity, Relationship, Asset	V1/V3/V4
TIM/VAL	F-TIM-*/F-VAL-*	REL-01/VAL-01	Event, ValueFlow	V1/V3/V4
TYP/HYP/ASM	F-TYP-*/F-HYP-*/ F-ASM-*	TYP-01/HYP-01/HYP-02/ ASM-01/GAP-01	Indicator, TypologyMatch, Hypothesis, Assessment	V1/V4/V6
PRD/REV/DIS	F-PRD-*/F-REV-001/ F-DIS-*	QUA-01/DIS-01/PRI-01	IntelligenceProduct, Review, Dissemination	V1/V2/V6
SEC/AUD/OPS	F-SEC-*/F-AUD-001/ F-OPS-001	SEC-01/SEC-02/AUD-01/ PRI-02	AccessPolicy, AuditEvent, BackupRecord	V2/V5

21. MVP Release Baseline

Release gate

MVP 0.1 SHALL NOT be declared production-ready until the end-to-end pilot scenario passes with provenance, authorization, audit, review, and export controls intact.

Non-waivable invariants [v0.1.1 · A16]

No administrative waiver is possible for a defect that violates: (1) authorization (unauthorized access / authorization bypass); (2) source identity protection; (3) evidence integrity or provenance of material records; (4) certainty preservation — e.g. RECONSTRUCTED/HYPOTHETICAL shown or stored as DIRECT/DOCUMENTED, INSUFFICIENT_BASIS shown as a level, or a claim treated as fact without a decision; (5) approval gates, including export without approval (SRS-FR-DIS-001); (6) audit history (broken, missing, or editable). If such a defect exists, the only release path is to disable the affected feature path with tested evidence of non-reachability; the defect itself is never waived. Other High defects may be waived only by the accountable authority with a tested compensating control, owner, and expiry date.

- A case can be opened only with accountable owner and valid charter.
- Evidence originals and derivatives are separable and integrity-protected.
- At least one entity merge/unmerge is demonstrated without history loss.
- Ownership/control/asset and mixed-class value flows are represented without semantic collapse.
- At least two competing hypotheses are tested using supporting and contradicting evidence.
- An assessment is confidence-rated and backward traceable to sources.
- A fact supported by a recorded source claim is created through a recorded verification decision (the claim is unchanged), and disputing/superseding that fact flags dependent assessments/products without loss of history. [v0.1.1 · A10]
- Independent peer review is completed.
- External export is blocked until dissemination approval.
- Audit trail captures all material actions.
- Backup restore drill succeeds in an isolated environment.

Annex A — Requirement ID Taxonomy

- SRS-FR-* Functional requirements
- SRS-DR-* Data requirements

- SRS-IF-* External interfaces
- SRS-SEC-* Security/privacy
- SRS-AUD-* Audit/provenance
- SRS-NFR-PERF-* Performance
- SRS-NFR-REL-* Reliability
- SRS-AN-* Search/graph/analytics
- SRS-AI-* AI/automation
- SRS-UX-* UX/accessibility
- SRS-ADM-* Administration
- SRS-INT-* Integration
- SRS-OPS-* Deployment/operations

Annex B — State Models

State values are the registered wire values of schemas/enums.yaml; the transitions follow the Data Model Specification v0.1.4 lifecycles. Labels such as "triage" or "changes requested" describe activities or review decisions, not stored states. [v0.1.1 · C08]

```
Case (case_status): DRAFT → AUTHORIZED → ACTIVE → REVIEW → CLOSED → MONITORING → REOPENED
Gate (gate_status): NOT_SUBMITTED → SUBMITTED → APPROVED | REJECTED; REJECTED → SUBMITTED
Task (task_status): OPEN | IN_PROGRESS | BLOCKED → DONE | CANCELLED (DONE, CANCELLED final)
DerivedText (derived_text_status): QUEUED → PROCESSING → REVIEW_REQUIRED → REVIEWED; QUEUED | PROCESSING → FAILED
```

Case status changes only through an approved gate: G0 DRAFT → AUTHORIZED, G1 AUTHORIZED → ACTIVE, G4 ACTIVE → REVIEW, G6 ACTIVE / REVIEW → CLOSED. MONITORING and REOPENED are not reachable through the API in v0.1.x (v0.2). [v0.1.4 · CR-N-03, CR-N-04, CR-N-09]

[v0.1.1 · C08] Per Data Model §6.1. Triage of an incoming matter happens while the case is DRAFT; authorization of purpose and scope moves it to AUTHORIZED. There is no APPROVED case state: approval applies to gates and products.

```
Entity resolution state (v0.1.1 · ER): UNRESOLVED → RESOLVED | CONFLICTED; absorbed record → MERGED; restored record → SPLIT
ResolutionDecision.decision (append-only): MERGE | KEEP_SEPARATE | POSSIBLE_MATCH | DEFER | UNMERGE
```

[v0.1.1 · ER] Entity state changes only as the effect of a ResolutionDecision (SRS-FR-ENT-006). v0.1 states CANDIDATE/PROBABLE are POSSIBLE_MATCH decisions; CONFIRMED → RESOLVED; DISPUTED → CONFLICTED.

```
Intelligence product (product_approval_state): DRAFT → REVIEWED → APPROVED → DISSEMINATED
DRAFT | REVIEWED | APPROVED | DISSEMINATED → WITHDRAWN
```

[v0.1.1 · C08] Per Data Model §15.1 (approval_state). A review decision RETURN (Data Model §15.2; "changes requested") sends the product back to DRAFT as a new version; corrections and supersession create a new product version (ST-E8-04) rather than a separate state. review_required (Section 7.5 of the Data Model) is a flag, not a state.

```
Hypothesis (hypothesis_status): OPEN → SUPPORTED | WEAKENED | REJECTED | INCONCLUSIVE
Claim (claim_status): RECORDED → UNDER_REVIEW → CORROBORATED | CONTRADICTED | UNRESOLVED
CORROBORATED | CONTRADICTED | UNRESOLVED → UNDER_REVIEW (re-review)
Fact (fact_status): PROVISIONAL → ESTABLISHED
PROVISIONAL | ESTABLISHED → DISPUTED
PROVISIONAL | ESTABLISHED | DISPUTED → SUPERSEDED (superseded_by required)
```

[v0.1.2 · CR-I2-11] A decided claim (CORROBORATED, CONTRADICTED, UNRESOLVED) MAY return to UNDER_REVIEW; the earlier decisions are kept.

```
Upload session (upload_session_status): INITIATED → CONTENT_RECEIVED → COMPLETED
Assessment (envelope status): DRAFT → FINALIZED → then follows review_status (PEER_REVIEWED | APPROVED | SUPERSEDED)
Indicator (indicator_status): OBSERVED → CORROBORATED | DISPUTED | RETIRED
DISPUTED → OBSERVED | CORROBORATED | RETIRED (RETIRED is terminal)
Lifecycle-less classes (envelope_status): Source, EvidenceExtract, Asset → REGISTERED; Event, ValueFlow, TypologyMatch → RECORDED
```

[v0.1.2 · CR-I2-02, CR-I2-03, CR-I3-05, CR-I4-02, CR-I4-09] An upload session that completed its content upload is either completed into an EvidenceItem or consumed by a derivative registration;

expiry (24 h) is expressed by `expires_at`, not a state. Assessment `review_status` stays DRAFT until a review outcome exists. Case memberships have no state model: a membership is active until its soft revocation (`revoked_at`). [v0.1.2 · CR-I1-02]

[v0.1.1 · A10] Claim/Fact states per SRS-FR-CLM-001...004. A Claim never transitions into a Fact. [v0.1.1 · C08] Fact transitions per Data Model §7.5, including PROVISIONAL → DISPUTED and DISPUTED → SUPERSEDED; every transition is recorded with its VerificationDecision.

Annex C — API Resource Baseline

Paths are relative to `/api/v1` and follow API Specification v0.1.4 §12–§20 and `contracts/openapi.yaml`. [v0.1.1 · C13]

- `/cases`, `/cases/{caseId}/charter`, `/cases/{caseId}/gates`, `/cases/{caseId}/tasks`
- `/cases/{caseId}/charter/versions`, `/cases/{caseId}/charter/versions/{version}`, `/tasks/{taskId}` [v0.1.4 · CR-N-01, CR-N-04]
- `/cases/{caseId}/memberships`, `/case-memberships/{membershipId}` [v0.1.2 · CR-I1-02]
- `/auth/backchannel-logout` (outside `/api/v1`, with the other `/auth/*` endpoints) [v0.1.2 · CR-I1-05]
- `/sources`
- `/evidence`, `/evidence/uploads`, `/evidence/{evidenceId}/extracts`, `/evidence/{evidenceId}/lineage` [v0.1.1 · C13]
- `/evidence/{evidenceId}/text-extractions`, `/evidence/{evidenceId}/translations`, `/evidence/{evidenceId}/derived-texts`, `/derived-texts/{derivedTextId}` (translations, review) [v0.1.4 · CR-N-09, CR-N-10]
- `/cases/{caseId}/claims`, `/claims/{claimId}`, `/claims/{claimId}/verification-decisions` [v0.1.1 · A10] [v0.1.1 · C13]
- `/cases/{caseId}/facts`, `/facts/{factId}` (commands: establish, dispute, supersede; dependents) [v0.1.1 · A10] [v0.1.1 · C13]; `/facts/{factId}/provenance` [v0.1.2 · CR-I2-04]
- `/entities`, `/entity-match-candidates`, `/entities/merge`, `/entity-merges/{mergelId}/unmerge`, `/entities/{entityId}/resolution-decisions`, `/resolution-decisions` [v0.1.1 · ER]; `/entities/{entityId}/provenance` [v0.1.2 · CR-I2-04]; `/entity-match-candidates/{candidateId}/decisions` is deprecated (removal in v0.2) — use `/resolution-decisions` [v0.1.2 · CR-I2-14]
- `/relationships`, `/relationships/{relationshipId}/provenance` [v0.1.2 · CR-I2-04]
- `/assets`, `/ownership-interests`, `/control-assertions` (each with `/id` and `/id/provenance`) [v0.1.2 · CR-I3-01, CR-I3-08]
- `/events`, `/events/{eventId}/provenance`, `/timeline` [v0.1.2 · CR-I3-02]
- `/value-flows`, `/value-flows/{flowId}/legs`, `/value-flows/{flowId}/provenance`, `/value-flow-view`, `/value-flow-legend` [v0.1.2 · CR-I3-03, CR-I3-08]
- `/cases/{caseId}/graph` [v0.1.2 · CR-I3-07]
- `/indicators`, `/indicators/{indicatorId}/provenance` [v0.1.2 · CR-I4-02, CR-I4-05]
- `/typology-catalogue`, `/typologies`, `/typologies/{typologyId}`, `/typology-matches`, `/typology-matches/evaluate`, `/typology-matches/{matchId}/provenance` [v0.1.2 · CR-I4-01, CR-I4-03, CR-I4-05]
- `/hypotheses`, `/hypotheses/{hypothesisId}/links`, `/hypotheses/{hypothesisId}/provenance`, `/hypothesis-matrix` [v0.1.2 · CR-I4-04, CR-I4-05]
- `/intelligence-gaps`, `/intelligence-gaps/{gapId}` [v0.1.1 · C13] [v0.1.2 · CR-I4-05]
- `/assessments` (commands: finalize, disconfirming-searches; provenance; revisions) [v0.1.1 · C13] [v0.1.2 · CR-I4-05]

- /products
- /reviews
- /disseminations, /sharing-log
- /audit-events
- /search, /graph/query

API rule This list is informative. The normative interface is `contracts/openapi.yaml` (SRS-IF-002), with semantics in the API Specification; where this list and those documents differ, they prevail. Authorization semantics, stable identifiers, provenance, and uncertainty preservation are normative. *[v0.1.1 · C13]*

Annex D — Test Scenario Baseline

1. Create a high-sensitivity case and charter.
2. Register two public sources and upload three evidence files; hash originals.
3. Create evidence extracts and one OCR derivative; record a source claim from an extract and create a PROVISIONAL fact supported by it via a verification decision. *[v0.1.1 · A10]*
4. Create two similar company records, evaluate candidate match, merge, then unmerge one test cycle; verify each step is recorded as a ResolutionDecision. *[v0.1.1 · ER]*
5. Create Person, Company, Property and Contract entities plus ownership/control relationships.
6. Add timeline events for incorporation, contract award, property acquisition and ownership change.
7. Create a four-leg value flow containing DIRECT, DOCUMENTED, RECONSTRUCTED and HYPOTHETICAL legs.
8. Map two indicators and one counter-indicator to a typology.
9. Create two competing hypotheses and record supporting/contradicting evidence.
10. Create assessment with MODERATE confidence and two intelligence gaps; record a disconfirming-search entry before review. *[v0.1.1 · A05, A09]*
11. Generate intelligence product, submit to independent reviewer, address change request, approve.
12. Attempt export before approval (must fail), then approve dissemination and export.
13. Verify sharing log and audit history.
14. Restore the case from backup into isolated environment and verify object/evidence hashes.